



## Handout 4.14: Model Build Guide and Label Cards

Print the build guide (pages 1 and 2) one per team. Print the label cards (page 3) on card stock, one set per team, and cut. Print the isometric grid page (page 4) for any team using the no-materials alternative.

## Page 1: The build guide, Day 1 (walls and floor)

Name(s): \_\_\_\_\_ Client: \_\_\_\_\_ Model scale (circle):  $1/2$  inch = 1 foot / 1 inch = 1 foot

**The scale.** A shoebox is about 13 inches by 8 inches inside. At  $1/2$  inch = 1 foot, that is a room 26 feet by 16 feet. So every client room fits at the same scale as your plan. If you use a bigger box or a foam board base, you may use 1 inch = 1 foot. Whichever you use, **write the scale on the rim of the box or the edge of the base.**

Picture 1: a shoebox with the lid off, the scale written on the rim.

**Step 1. Mark the floor.** Find the room's length and width on your plan. Count the squares. Change squares to feet. Change feet to model inches. Draw the room's outline on the floor of the box with a pencil and ruler. If the room is smaller than the box, the extra floor is outside the room. Add card walls later, or just mark the line. Tiny house: at  $1/2$  inch = 1 foot, the floor is 4 inches by 12 inches (8 by 24 feet). Mark it.

Picture 2: a ruler on the box floor with a pencil line.

**Step 2. Mark the door and windows.** Find the wall and the spot from your plan. Draw the door opening (3 feet is 1.5 inches at  $1/2$  inch = 1 foot; 3 inches at 1 inch = 1 foot). Take the box to the teacher's cutting station to cut the door opening. Draw the window as a frame with a marker, or cut it out at the station.

Picture 3: a box wall with a door cut out and a window drawn.

**Step 3. Extra walls.** The closet: cut a card strip as deep as the closet. Fold a tab at the bottom. Glue it in. The tiny house bathroom: two card walls, 3 by 5 feet at scale, with a gap for the door. The tiny house loft: a card shelf glued across the end of the box, 4 feet below the rim (2 inches at  $1/2$  inch = 1 foot). Wall height: rooms are 8 feet. That is 4 inches at  $1/2$  inch = 1 foot. Most shoeboxes are 4 to 5 inches deep, so the box wall is the room wall.

Picture 4: a card strip folded into an L, glued to the floor as a closet wall.

**Step 4. The floor covering.** Cut paper, felt, fabric, or a printed wood or tile pattern to the room's floor size and glue it down. A rug (if the client allows one) goes on later, as its own piece.

Picture 5: felt glued to the box floor.

**Station check before the bell:** glue caps on, scraps in the scrap bin (not the trash), model on the shelf with your team name on masking tape.

## Page 2: The build guide, Day 2 (color and furniture) and Day 3 (finish and labels)

**Day 2, Step 5. Color first.** Pull your Lesson 4.8 color card. Cover the walls with the main color (paper, a paint chip, marker, or fabric). The accent color goes on the focal point: the headboard, a rug, a chair, or one wall. Later, name the scheme (monochromatic, analogous, complementary) on the label. Do the walls before the furniture goes in. You cannot paint behind a bed.

**Texture, too.** Pull your Lesson 4.9 mood board. Put one texture from it on the model: a fabric scrap on the bed, a rough paper on one wall, or a cork square for a rug. Rubric 04 looks for one texture.

Picture 6: three walls covered in light blue paper, one wall a darker blue.

**Day 2, Step 6. Furniture from card and scrap, to scale.** Every piece on your plan gets built. Use the cut-out sheet sizes (Handout 04.12, page 2). Convert: at 1/2 inch = 1 foot, a 3 by 6.5 foot twin bed is 1.5 by 3.25 inches. Ways to build:

| Piece                    | How                                                                                                                                   |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Bed or crib              | A small card box (fold a rectangle, tab the ends) with a fabric scrap as the blanket. A crib gets card rails.                         |
| Desk or table            | A card rectangle with four legs (toothpicks, rolled paper, or a folded card skirt). A wheelchair user's desk has open space under it. |
| Chair or loveseat        | A card L (seat and back) with a folded card base, or a foam scrap. Arms are two small card strips.                                    |
| Dresser, bookcase, cubes | A stacked card block. Draw the drawers or shelves on the front.                                                                       |
| Wardrobe or closet rod   | A card box with a toothpick rod.                                                                                                      |
| Lamp                     | A toothpick or a paper straw with a paper cone.                                                                                       |
| Rug                      | A fabric or felt scrap, cut to the plan's size, taped down (the walker and wheelchair rule).                                          |
| Window covering          | A fabric strip glued above the window, or a paper blind. No dangling strings on the Ahmed model.                                      |
| Hooks                    | Pins or bent paper clips in the wall at two heights.                                                                                  |
| Plant                    | A green paper scrap in a card cup.                                                                                                    |
| Tiny house kitchen       | A card block with a drawn cooktop and a                                                                                               |

|                     |                                                            |
|---------------------|------------------------------------------------------------|
|                     | foil sink. The fridge is a card box.                       |
| Tiny house bathroom | A card shower base and a card toilet inside the two walls. |
| Loft ladder         | Toothpicks or card strips glued in a ladder.               |

Place every piece where the plan has it. Hold the plan up next to the box. **Do not glue a piece down until it matches the plan.** If a piece looks better somewhere else, that is allowed; change the plan too (in pencil, with a note).

Picture 7: a card bed with a fabric blanket next to the plan showing the same bed on the same wall.

**Day 3, Step 7. Finish.** Glue the furniture. Add the lamp, the rug, the window covering, the plant, the hooks. Fix anything out of scale.

**Day 3, Step 8. The labels.** Fill the three cards on page 3 and glue them to the inside wall or the rim:

- GREEN: one green choice from Lesson 4.5 and why it matters for this client.
- SAFE: one safety feature from Lesson 4.2 and why it matters for this client.
- WHY: two sentences. Sentence one: the biggest design choice and the client need it serves. Sentence two: the principle of design it uses (balance, focal point, rhythm, proportion, unity, contrast).

**Day 3, Step 9. The client card** goes on the outside of the box, front wall. **The scale** is on the rim. **Your names** on masking tape on the bottom.

**Day 3, Step 10. Self-check.** Read Rubric 04, criterion 4. Circle your level. Fix one thing.

Picture 8: the finished model with the three label cards inside and the client card on the front.

Word bank: wall (pared), floor (piso), door (puerta), window (ventana), glue (pegar), cut (cortar), label (etiqueta).

### Page 3: Label cards (cut on the lines)

#### GREEN

This is: \_\_\_\_\_

It is a green choice because it \_\_\_\_\_ (conserves energy / conserves water / reuses something / reduces waste / cleaner air).

It matters for this client because \_\_\_\_\_.

Frame: "This is . ***It is green because the client .***"

#### SAFE

This is: \_\_\_\_\_

It is a safety feature because it prevents \_\_\_\_\_ (a fall / a trip / a burn / a cord hazard / a crib hazard / a blocked exit).

It matters for this client because \_\_\_\_\_.

Frame: "This is . ***It is safe because the client .***"

#### WHY (the justification)

The biggest design choice in this room is \_\_\_\_\_, and it serves the client's need for \_\_\_\_\_.

It uses the principle of \_\_\_\_\_ (balance / focal point / rhythm / proportion / unity / contrast) because \_\_\_\_\_.

#### ACCESSIBLE (grade 8 stretch, optional)

This is: \_\_\_\_\_ It is set at \_\_\_\_\_ inches so that \_\_\_\_\_.

#### Client card mount

Glue the client card here (outside, front wall). Scale of this model: \_\_\_\_\_ inch = 1 foot.

## Page 4: The isometric grid alternative (no box, no materials, or absent for the build days)

Draw the room in 3-D on isometric grid paper (the triangle grid). If the school has none, print a free isometric sheet or draw the room on plain paper using the steps below.

**Scale:** 1/2 inch = 1 foot. One grid triangle side is 1/4 inch, so two triangle sides equal 1 foot.

1. **The floor.** Draw the room's floor as a parallelogram: the length along one grid direction, the width along the other, counting two triangle sides per foot.
2. **The walls.** From the two back corners and the middle back corner, draw straight up 8 feet (16 triangle sides). Connect the tops. You now have a floor and two back walls, like an open box seen from a corner.
3. **The door and window.** Draw the door as an opening in the wall the plan puts it on, 3 feet wide and 7 feet tall. Draw the window as a rectangle in its wall at the plan's spot, its bottom 3 feet off the floor.
4. **The furniture.** Every piece on the plan is a box: draw its footprint on the floor where the plan has it (count the squares from the plan; two triangle sides per foot), then draw it up to its height. Typical heights: bed 2 feet, crib 3 feet, desk 2.5 feet, chair seat 1.5 feet with a back to 3 feet, dresser 4 feet, bookcase 3.5 feet, wardrobe 6 feet, table 2.5 feet.
5. **The color.** Color the walls in the main color, the focal point in the accent. Name the scheme in the corner.
6. **The labels.** Draw the GREEN and SAFE labels as callouts with an arrow to the feature. Write the WHY sentences under the drawing.
7. **The client card** is stapled to the drawing. **The scale** is written in the corner.

You may also build a digital 3-D room in any free room-planner app the school allows. Print it, or show it on a screen at the gallery walk. It needs the same labels, the same scale note, and the client card.

It is scored on Rubric 04 criterion 4 exactly like a box model.